RETAINED GLOSSARY

Technical terms to keep

A ranked, index-grounded vocabulary checklist for Build a Large Language Model (From Scratch).

265 terms	537 index entries checked	4 importance tiers

Importance is editorial study guidance, not a rating supplied by the textbook. ESSENTIAL = foundational; HIGH = practically important; SPECIALIZED = narrower technique or implementation detail; PERIPHERAL = legitimate but incidental terminology. Every ranked row explicitly labels its keyword, topic and importance.

01 / ESSENTIAL - 54 terms

RANK	TECHNICAL TERM DETAILS
001	**Keyword:** Large language models (LLMs) **Topic:** LLM Foundations **Importance:** Essential
002	**Keyword:** Generative Pre-trained Transformer (GPT) **Topic:** LLM Foundations **Importance:** Essential
003	**Keyword:** Transformer architecture **Topic:** LLM Foundations **Importance:** Essential
004	**Keyword:** Generative artificial intelligence (GenAI) **Topic:** LLM Foundations **Importance:** Essential
005	**Keyword:** Autoregressive language modeling **Topic:** LLM Foundations **Importance:** Essential
006	**Keyword:** Pretraining **Topic:** LLM Foundations **Importance:** Essential

RANK	TECHNICAL TERM DETAILS
007	**Keyword:** Fine-tuning **Topic:** LLM Foundations **Importance:** Essential
008	**Keyword:** Foundation models **Topic:** LLM Foundations **Importance:** Essential
009	**Keyword:** Language model training objectives **Topic:** LLM Foundations **Importance:** Essential
010	**Keyword:** Tokenization **Topic:** Text Processing **Importance:** Essential
011	**Keyword:** Byte pair encoding (BPE) **Topic:** Text Processing **Importance:** Essential
012	**Keyword:** Token IDs **Topic:** Text Processing **Importance:** Essential
013	**Keyword:** Token embeddings **Topic:** Text Processing **Importance:** Essential
014	**Keyword:** Word embeddings **Topic:** Text Processing **Importance:** Essential
015	**Keyword:** Positional embeddings **Topic:** Text Processing **Importance:** Essential
016	**Keyword:** Context length **Topic:** Text Processing **Importance:** Essential

RANK	TECHNICAL TERM DETAILS
017	**Keyword:** Special context tokens **Topic:** Text Processing **Importance:** Essential
018	**Keyword:** Attention mechanisms **Topic:** Transformer Attention **Importance:** Essential
019	**Keyword:** Self-attention **Topic:** Transformer Attention **Importance:** Essential
020	**Keyword:** Causal attention **Topic:** Transformer Attention **Importance:** Essential
021	**Keyword:** Multi-head attention **Topic:** Transformer Attention **Importance:** Essential
022	**Keyword:** Attention scores **Topic:** Transformer Attention **Importance:** Essential
023	**Keyword:** Attention weights **Topic:** Transformer Attention **Importance:** Essential
024	**Keyword:** Context vectors **Topic:** Transformer Attention **Importance:** Essential
025	**Keyword:** Causal attention masks **Topic:** Transformer Attention **Importance:** Essential
026	**Keyword:** Scaled dot-product attention **Topic:** Transformer Attention **Importance:** Essential

RANK	TECHNICAL TERM DETAILS
027	**Keyword:** Transformer blocks **Topic:** Model Architecture **Importance:** Essential
028	**Keyword:** Feedforward neural networks **Topic:** Model Architecture **Importance:** Essential
029	**Keyword:** Layer normalization **Topic:** Model Architecture **Importance:** Essential
030	**Keyword:** Shortcut / residual connections **Topic:** Model Architecture **Importance:** Essential
031	**Keyword:** Linear / fully connected layers **Topic:** Model Architecture **Importance:** Essential
032	**Keyword:** Activation functions **Topic:** Model Architecture **Importance:** Essential
033	**Keyword:** Softmax **Topic:** Model Architecture **Importance:** Essential
034	**Keyword:** Tensors **Topic:** Training & Evaluation **Importance:** Essential
035	**Keyword:** Automatic differentiation (autograd) **Topic:** Training & Evaluation **Importance:** Essential
036	**Keyword:** Backpropagation **Topic:** Training & Evaluation **Importance:** Essential

RANK	TECHNICAL TERM DETAILS
037	**Keyword:** Gradients **Topic:** Training & Evaluation **Importance:** Essential
038	**Keyword:** Loss functions **Topic:** Training & Evaluation **Importance:** Essential
039	**Keyword:** Cross-entropy loss **Topic:** Training & Evaluation **Importance:** Essential
040	**Keyword:** Training loops **Topic:** Training & Evaluation **Importance:** Essential
041	**Keyword:** Training and validation losses **Topic:** Training & Evaluation **Importance:** Essential
042	**Keyword:** Optimization algorithms **Topic:** Training & Evaluation **Importance:** Essential
043	**Keyword:** Learning rate **Topic:** Training & Evaluation **Importance:** Essential
044	**Keyword:** Text generation **Topic:** Text Generation **Importance:** Essential
045	**Keyword:** Decoding strategies **Topic:** Text Generation **Importance:** Essential
046	**Keyword:** Greedy decoding **Topic:** Text Generation **Importance:** Essential

RANK	TECHNICAL TERM DETAILS
047	**Keyword:** Temperature scaling **Topic:** Text Generation **Importance:** Essential
048	**Keyword:** Top-k sampling **Topic:** Text Generation **Importance:** Essential
049	**Keyword:** Instruction fine-tuning **Topic:** Fine-tuning & Adaptation **Importance:** Essential
050	**Keyword:** Supervised fine-tuning **Topic:** Fine-tuning & Adaptation **Importance:** Essential
051	**Keyword:** Classification fine-tuning **Topic:** Fine-tuning & Adaptation **Importance:** Essential
052	**Keyword:** Parameter-efficient fine-tuning (PEFT) **Topic:** Fine-tuning & Adaptation **Importance:** Essential
053	**Keyword:** Low-rank adaptation (LoRA) **Topic:** Fine-tuning & Adaptation **Importance:** Essential
054	**Keyword:** Preference fine-tuning **Topic:** Fine-tuning & Adaptation **Importance:** Essential

02 / HIGH - 105 terms

RANK	TECHNICAL TERM DETAILS
055	**Keyword:** Artificial intelligence (AI) **Topic:** LLM Foundations **Importance:** High

RANK	TECHNICAL TERM DETAILS
056	**Keyword:** Machine learning **Topic:** LLM Foundations **Importance:** High
057	**Keyword:** Deep learning **Topic:** LLM Foundations **Importance:** High
058	**Keyword:** Supervised learning **Topic:** LLM Foundations **Importance:** High
059	**Keyword:** Unlabeled data pretraining **Topic:** LLM Foundations **Importance:** High
060	**Keyword:** Emergent behavior **Topic:** LLM Foundations **Importance:** High
061	**Keyword:** Large-scale pretraining datasets **Topic:** LLM Foundations **Importance:** High
062	**Keyword:** Model architecture implementation **Topic:** LLM Foundations **Importance:** High
063	**Keyword:** Model parameters **Topic:** LLM Foundations **Importance:** High
064	**Keyword:** Model weights **Topic:** LLM Foundations **Importance:** High
065	**Keyword:** Pretrained weights **Topic:** LLM Foundations **Importance:** High

RANK	TECHNICAL TERM DETAILS
066	**Keyword:** Vocabulary size **Topic:** Text Processing **Importance:** High
067	**Keyword:** Token embedding matrices **Topic:** Text Processing **Importance:** High
068	**Keyword:** Embedding dimension **Topic:** Text Processing **Importance:** High
069	**Keyword:** Sliding-window sampling **Topic:** Text Processing **Importance:** High
070	**Keyword:** Input/output token sequences **Topic:** Text Processing **Importance:** High
071	**Keyword:** Instruction-response pairs **Topic:** Text Processing **Importance:** High
072	**Keyword:** Prompt formatting **Topic:** Text Processing **Importance:** High
073	**Keyword:** Text completion **Topic:** Text Processing **Importance:** High
074	**Keyword:** Encoding and decoding tokens **Topic:** Text Processing **Importance:** High
075	**Keyword:** Raw text preprocessing **Topic:** Text Processing **Importance:** High

RANK	TECHNICAL TERM DETAILS
076	**Keyword:** Query, key and value matrices **Topic:** Transformer Attention **Importance:** High
077	**Keyword:** Dot products **Topic:** Transformer Attention **Importance:** High
078	**Keyword:** Trainable attention weights **Topic:** Transformer Attention **Importance:** High
079	**Keyword:** Multi-head weight splitting **Topic:** Transformer Attention **Importance:** High
080	**Keyword:** Stacking single-head attention layers **Topic:** Transformer Attention **Importance:** High
081	**Keyword:** Preventing information leakage **Topic:** Transformer Attention **Importance:** High
082	**Keyword:** Long-sequence modeling **Topic:** Transformer Attention **Importance:** High
083	**Keyword:** Single-head attention **Topic:** Transformer Attention **Importance:** High
084	**Keyword:** Number of attention heads **Topic:** Transformer Attention **Importance:** High
085	**Keyword:** GELU activation **Topic:** Model Architecture **Importance:** High

RANK	TECHNICAL TERM DETAILS
086	**Keyword:** ReLU activation **Topic:** Model Architecture **Importance:** High
087	**Keyword:** SwiGLU activation **Topic:** Model Architecture **Importance:** High
088	**Keyword:** Dropout regularization **Topic:** Model Architecture **Importance:** High
089	**Keyword:** Pre-LayerNorm **Topic:** Model Architecture **Importance:** High
090	**Keyword:** Post-LayerNorm **Topic:** Model Architecture **Importance:** High
091	**Keyword:** RMSNorm **Topic:** Model Architecture **Importance:** High
092	**Keyword:** Embedding layer **Topic:** Model Architecture **Importance:** High
093	**Keyword:** Output / classification heads **Topic:** Model Architecture **Importance:** High
094	**Keyword:** Forward pass **Topic:** Model Architecture **Importance:** High
095	**Keyword:** Linear layer weights **Topic:** Model Architecture **Importance:** High

RANK	TECHNICAL TERM DETAILS
096	**Keyword:** Multilayer neural networks **Topic:** Model Architecture **Importance:** High
097	**Keyword:** Computation graphs **Topic:** Training & Evaluation **Importance:** High
098	**Keyword:** Partial derivatives **Topic:** Training & Evaluation **Importance:** High
099	**Keyword:** Scalar, vector and matrix operations **Topic:** Training & Evaluation **Importance:** High
100	**Keyword:** Matrix multiplication **Topic:** Training & Evaluation **Importance:** High
101	**Keyword:** Tensor data types **Topic:** Training & Evaluation **Importance:** High
102	**Keyword:** Tensor shapes and dimensions **Topic:** Training & Evaluation **Importance:** High
103	**Keyword:** Data loaders **Topic:** Training & Evaluation **Importance:** High
104	**Keyword:** Datasets and dataset classes **Topic:** Training & Evaluation **Importance:** High
105	**Keyword:** Training batches **Topic:** Training & Evaluation **Importance:** High

RANK	TECHNICAL TERM DETAILS
106	**Keyword:** Stochastic gradient descent (SGD) **Topic:** Training & Evaluation **Importance:** High
107	**Keyword:** AdamW optimizer **Topic:** Training & Evaluation **Importance:** High
108	**Keyword:** Learning rate warmup **Topic:** Training & Evaluation **Importance:** High
109	**Keyword:** Cosine learning-rate decay **Topic:** Training & Evaluation **Importance:** High
110	**Keyword:** Gradient clipping **Topic:** Training & Evaluation **Importance:** High
111	**Keyword:** Weight decay **Topic:** Training & Evaluation **Importance:** High
112	**Keyword:** Model evaluation **Topic:** Training & Evaluation **Importance:** High
113	**Keyword:** Accuracy metric **Topic:** Training & Evaluation **Importance:** High
114	**Keyword:** Perplexity **Topic:** Training & Evaluation **Importance:** High
115	**Keyword:** Model checkpointing **Topic:** Training & Evaluation **Importance:** High

RANK	TECHNICAL TERM DETAILS
116	**Keyword:** Training / evaluation modes **Topic:** Training & Evaluation **Importance:** High
117	**Keyword:** Sampling from model logits **Topic:** Text Generation **Importance:** High
118	**Keyword:** Multinomial sampling **Topic:** Text Generation **Importance:** High
119	**Keyword:** Randomness control in decoding **Topic:** Text Generation **Importance:** High
120	**Keyword:** Model text-generation functions **Topic:** Text Generation **Importance:** High
121	**Keyword:** Sequence length constraints **Topic:** Text Generation **Importance:** High
122	**Keyword:** Instruction datasets **Topic:** Fine-tuning & Adaptation **Importance:** High
123	**Keyword:** Instruction-following models **Topic:** Fine-tuning & Adaptation **Importance:** High
124	**Keyword:** Classification heads **Topic:** Fine-tuning & Adaptation **Importance:** High
125	**Keyword:** Supervised classification loss **Topic:** Fine-tuning & Adaptation **Importance:** High

RANK	TECHNICAL TERM DETAILS
126	**Keyword:** Spam classification with LLMs **Topic:** Fine-tuning & Adaptation **Importance:** High
127	**Keyword:** Instruction-data preparation **Topic:** Fine-tuning & Adaptation **Importance:** High
128	**Keyword:** Instruction-batch collation **Topic:** Fine-tuning & Adaptation **Importance:** High
129	**Keyword:** Fine-tuned response evaluation **Topic:** Fine-tuning & Adaptation **Importance:** High
130	**Keyword:** Low-rank scaling factor **Topic:** Fine-tuning & Adaptation **Importance:** High
131	**Keyword:** LoRA adapted linear layers **Topic:** Fine-tuning & Adaptation **Importance:** High
132	**Keyword:** Loading pretrained LLMs **Topic:** Fine-tuning & Adaptation **Importance:** High
133	**Keyword:** Pretrained weight initialization **Topic:** Fine-tuning & Adaptation **Importance:** High
134	**Keyword:** GPU-accelerated training **Topic:** Performance & Distributed Training **Importance:** High
135	**Keyword:** Single-GPU training **Topic:** Performance & Distributed Training **Importance:** High

RANK	TECHNICAL TERM DETAILS
136	**Keyword:** Multi-GPU training **Topic:** Performance & Distributed Training **Importance:** High
137	**Keyword:** DistributedDataParallel (DDP) **Topic:** Performance & Distributed Training **Importance:** High
138	**Keyword:** Distributed data sampling **Topic:** Performance & Distributed Training **Importance:** High
139	**Keyword:** GPU device selection **Topic:** Performance & Distributed Training **Importance:** High
140	**Keyword:** Computations on GPUs **Topic:** Performance & Distributed Training **Importance:** High
141	**Keyword:** Gradient accumulation and updates **Topic:** Performance & Distributed Training **Importance:** High
142	**Keyword:** Gradient norm **Topic:** Performance & Distributed Training **Importance:** High
143	**Keyword:** PyTorch **Topic:** Python & PyTorch **Importance:** High
144	**Keyword:** Automatic gradient tracking **Topic:** Python & PyTorch **Importance:** High
145	**Keyword:** PyTorch Dataset abstraction **Topic:** Python & PyTorch **Importance:** High

RANK	TECHNICAL TERM DETAILS
146	**Keyword:** PyTorch DataLoader abstraction **Topic:** Python & PyTorch **Importance:** High
147	**Keyword:** PyTorch nn.Module abstraction **Topic:** Python & PyTorch **Importance:** High
148	**Keyword:** PyTorch Linear layer **Topic:** Python & PyTorch **Importance:** High
149	**Keyword:** Tensor operations **Topic:** Python & PyTorch **Importance:** High
150	**Keyword:** State dictionaries **Topic:** Python & PyTorch **Importance:** High
151	**Keyword:** Torch no-gradient context **Topic:** Python & PyTorch **Importance:** High
152	**Keyword:** PyTorch model saving and loading **Topic:** Python & PyTorch **Importance:** High
153	**Keyword:** PyTorch tensor creation **Topic:** Python & PyTorch **Importance:** High
154	**Keyword:** PyTorch random seeds **Topic:** Python & PyTorch **Importance:** High
155	**Keyword:** Base model **Topic:** LLM Foundations **Importance:** High

RANK	TECHNICAL TERM DETAILS
156	**Keyword:** Neural networks **Topic:** LLM Foundations **Importance:** High
157	**Keyword:** Encoder and decoder architectures **Topic:** Model Architecture **Importance:** High
158	**Keyword:** Feedforward layer **Topic:** Model Architecture **Importance:** High
159	**Keyword:** Tokenizers **Topic:** Text Processing **Importance:** High

03 / SPECIALIZED - 91 terms

RANK	TECHNICAL TERM DETAILS
160	**Keyword:** BERT **Topic:** Language Models & Tools **Importance:** Specialized
161	**Keyword:** GPT-2 **Topic:** Language Models & Tools **Importance:** Specialized
162	**Keyword:** GPT-3 **Topic:** Language Models & Tools **Importance:** Specialized
163	**Keyword:** GPT-4 **Topic:** Language Models & Tools **Importance:** Specialized
164	**Keyword:** Llama 2 **Topic:** Language Models & Tools **Importance:** Specialized

RANK	TECHNICAL TERM DETAILS
165	**Keyword:** Llama 3 **Topic:** Language Models & Tools **Importance:** Specialized
166	**Keyword:** Phi-3 **Topic:** Language Models & Tools **Importance:** Specialized
167	**Keyword:** OLMo **Topic:** Language Models & Tools **Importance:** Specialized
168	**Keyword:** Recurrent neural networks (RNNs) **Topic:** Language Models & Tools **Importance:** Specialized
169	**Keyword:** Bahdanau attention **Topic:** Language Models & Tools **Importance:** Specialized
170	**Keyword:** Word2Vec **Topic:** Language Models & Tools **Importance:** Specialized
171	**Keyword:** Retrieval-augmented generation (RAG) **Topic:** Language Models & Tools **Importance:** Specialized
172	**Keyword:** Ollama **Topic:** Language Models & Tools **Importance:** Specialized
173	**Keyword:** llama.cpp **Topic:** Language Models & Tools **Importance:** Specialized
174	**Keyword:** LitGPT **Topic:** Language Models & Tools **Importance:** Specialized

RANK	TECHNICAL TERM DETAILS
175	**Keyword:** Axolotl **Topic:** Language Models & Tools **Importance:** Specialized
176	**Keyword:** tiktoken **Topic:** Language Models & Tools **Importance:** Specialized
177	**Keyword:** Google Colab **Topic:** Language Models & Tools **Importance:** Specialized
178	**Keyword:** Python package installer (pip) **Topic:** Language Models & Tools **Importance:** Specialized
179	**Keyword:** Alpaca dataset **Topic:** Language Models & Tools **Importance:** Specialized
180	**Keyword:** FineWeb dataset **Topic:** Language Models & Tools **Importance:** Specialized
181	**Keyword:** LIMA dataset **Topic:** Language Models & Tools **Importance:** Specialized
182	**Keyword:** UltraChat dataset **Topic:** Language Models & Tools **Importance:** Specialized
183	**Keyword:** Dolma corpus **Topic:** Language Models & Tools **Importance:** Specialized
184	**Keyword:** Tensor scalar (0D) **Topic:** Python & PyTorch **Importance:** Specialized

RANK	TECHNICAL TERM DETAILS
185	**Keyword:** Tensor vector (1D) **Topic:** Python & PyTorch **Importance:** Specialized
186	**Keyword:** Two-dimensional tensors **Topic:** Python & PyTorch **Importance:** Specialized
187	**Keyword:** Three-dimensional tensors **Topic:** Python & PyTorch **Importance:** Specialized
188	**Keyword:** Tensor indexing and reshaping **Topic:** Python & PyTorch **Importance:** Specialized
189	**Keyword:** Tensor transposition **Topic:** Python & PyTorch **Importance:** Specialized
190	**Keyword:** Tensor casting and device transfer **Topic:** Python & PyTorch **Importance:** Specialized
191	**Keyword:** Model parameter inspection **Topic:** Python & PyTorch **Importance:** Specialized
192	**Keyword:** PyTorch training step **Topic:** Python & PyTorch **Importance:** Specialized
193	**Keyword:** PyTorch zero_grad **Topic:** Python & PyTorch **Importance:** Specialized
194	**Keyword:** Sampling API torch.argmax **Topic:** Python & PyTorch **Importance:** Specialized

RANK	TECHNICAL TERM DETAILS
195	**Keyword:** NumPy-like array API **Topic:** Python & PyTorch **Importance:** Specialized
196	**Keyword:** Tensor shapes **Topic:** Python & PyTorch **Importance:** Specialized
197	**Keyword:** Batch normalization **Topic:** Training & Evaluation **Importance:** Specialized
198	**Keyword:** Model initialization **Topic:** Training & Evaluation **Importance:** Specialized
199	**Keyword:** Batch size **Topic:** Training & Evaluation **Importance:** Specialized
200	**Keyword:** Target tensors **Topic:** Training & Evaluation **Importance:** Specialized
201	**Keyword:** Dataset train/test splits **Topic:** Training & Evaluation **Importance:** Specialized
202	**Keyword:** Test-set evaluation **Topic:** Training & Evaluation **Importance:** Specialized
203	**Keyword:** Weight initialization from pretraining **Topic:** Training & Evaluation **Importance:** Specialized
204	**Keyword:** Training step visualization **Topic:** Training & Evaluation **Importance:** Specialized

RANK	TECHNICAL TERM DETAILS
205	**Keyword:** Learning-rate schedule peak **Topic:** Training & Evaluation **Importance:** Specialized
206	**Keyword:** DataLoader batching boundaries **Topic:** Training & Evaluation **Importance:** Specialized
207	**Keyword:** Training ratio **Topic:** Training & Evaluation **Importance:** Specialized
208	**Keyword:** Model output logits **Topic:** Training & Evaluation **Importance:** Specialized
209	**Keyword:** Neural network parameters **Topic:** Training & Evaluation **Importance:** Specialized
210	**Keyword:** LoRA parameterization **Topic:** Fine-tuning & Adaptation **Importance:** Specialized
211	**Keyword:** LoRA layer replacement **Topic:** Fine-tuning & Adaptation **Importance:** Specialized
212	**Keyword:** Direct preference optimization (DPO) example folder **Topic:** Fine-tuning & Adaptation **Importance:** Specialized
213	**Keyword:** Instruction prompt templates **Topic:** Fine-tuning & Adaptation **Importance:** Specialized
214	**Keyword:** Instruction dataset collation **Topic:** Fine-tuning & Adaptation **Importance:** Specialized

RANK	TECHNICAL TERM DETAILS
215	**Keyword:** Instruction response extraction **Topic:** Fine-tuning & Adaptation **Importance:** Specialized
216	**Keyword:** Classification accuracy evaluation **Topic:** Fine-tuning & Adaptation **Importance:** Specialized
217	**Keyword:** PyTorch multi-process spawning **Topic:** Performance & Distributed Training **Importance:** Specialized
218	**Keyword:** Distributed process-group initialization **Topic:** Performance & Distributed Training **Importance:** Specialized
219	**Keyword:** Distributed process-group destruction **Topic:** Performance & Distributed Training **Importance:** Specialized
220	**Keyword:** CUDA device visibility **Topic:** Performance & Distributed Training **Importance:** Specialized
221	**Keyword:** Apple MPS device **Topic:** Performance & Distributed Training **Importance:** Specialized
222	**Keyword:** Distributed sampler rank **Topic:** Performance & Distributed Training **Importance:** Specialized
223	**Keyword:** Progress monitoring **Topic:** Performance & Distributed Training **Importance:** Specialized
224	**Keyword:** PyTorch multiprocessing **Topic:** Performance & Distributed Training **Importance:** Specialized

RANK	TECHNICAL TERM DETAILS
225	**Keyword:** Start-of-sequence token [BOS] **Topic:** Text Processing **Importance:** Specialized
226	**Keyword:** End-of-sequence token [EOS] **Topic:** Text Processing **Importance:** Specialized
227	**Keyword:** Padding token [PAD] **Topic:** Text Processing **Importance:** Specialized
228	**Keyword:** End-of-text token **Topic:** Text Processing **Importance:** Specialized
229	**Keyword:** Unknown token **Topic:** Text Processing **Importance:** Specialized
230	**Keyword:** Sequence padding and truncation **Topic:** Text Processing **Importance:** Specialized
231	**Keyword:** Tokenizer configuration **Topic:** Text Processing **Importance:** Specialized
232	**Keyword:** Text sequence stride **Topic:** Text Processing **Importance:** Specialized
233	**Keyword:** QKV projection bias **Topic:** Text Processing **Importance:** Specialized
234	**Keyword:** Context length parameterization **Topic:** Text Processing **Importance:** Specialized

RANK	TECHNICAL TERM DETAILS
235	**Keyword:** Attention weight projections **Topic:** Transformer Attention **Importance:** Specialized
236	**Keyword:** Attention head dimensions **Topic:** Transformer Attention **Importance:** Specialized
237	**Keyword:** Causal mask triangular construction **Topic:** Transformer Attention **Importance:** Specialized
238	**Keyword:** Attention output projection **Topic:** Transformer Attention **Importance:** Specialized
239	**Keyword:** Single-head attention implementation **Topic:** Transformer Attention **Importance:** Specialized
240	**Keyword:** Causal-attention implementation **Topic:** Transformer Attention **Importance:** Specialized
241	**Keyword:** Multi-head attention implementation **Topic:** Transformer Attention **Importance:** Specialized
242	**Keyword:** GELU feedforward implementation **Topic:** Model Architecture **Importance:** Specialized
243	**Keyword:** GPT model implementation **Topic:** Model Architecture **Importance:** Specialized
244	**Keyword:** Transformer block implementation **Topic:** Model Architecture **Importance:** Specialized

RANK	TECHNICAL TERM DETAILS
245	**Keyword:** LayerNorm module **Topic:** Model Architecture **Importance:** Specialized
246	**Keyword:** RMSNorm alternative **Topic:** Model Architecture **Importance:** Specialized
247	**Keyword:** Sampling argmax **Topic:** Text Generation **Importance:** Specialized
248	**Keyword:** Model response generation **Topic:** Text Generation **Importance:** Specialized
249	**Keyword:** Logistic regression loss **Topic:** Training & Evaluation **Importance:** Specialized
250	**Keyword:** Conversational performance **Topic:** Fine-tuning & Adaptation **Importance:** Specialized

04 / PERIPHERAL - 15 terms

RANK	TECHNICAL TERM DETAILS
251	**Keyword:** PHUDGE model **Topic:** Language Models & Tools **Importance:** Peripheral
252	**Keyword:** Prometheus model **Topic:** Language Models & Tools **Importance:** Peripheral
253	**Keyword:** Python regular expressions **Topic:** Language Models & Tools **Importance:** Peripheral

RANK	TECHNICAL TERM DETAILS
254	**Keyword:** Python DataFrame **Topic:** Language Models & Tools **Importance:** Peripheral
255	**Keyword:** Python code timing **Topic:** Language Models & Tools **Importance:** Peripheral
256	**Keyword:** PyTorch audio library **Topic:** Language Models & Tools **Importance:** Peripheral
257	**Keyword:** PyTorch vision library **Topic:** Language Models & Tools **Importance:** Peripheral
258	**Keyword:** Python version **Topic:** Language Models & Tools **Importance:** Peripheral
259	**Keyword:** macOS **Topic:** Language Models & Tools **Importance:** Peripheral
260	**Keyword:** Math equality operator **Topic:** Language Models & Tools **Importance:** Peripheral
261	**Keyword:** Python matrix multiply operator **Topic:** Language Models & Tools **Importance:** Peripheral
262	**Keyword:** Tensor flattening / element count **Topic:** Language Models & Tools **Importance:** Peripheral
263	**Keyword:** Python functools library **Topic:** Language Models & Tools **Importance:** Peripheral

RANK	TECHNICAL TERM DETAILS
264	**Keyword:** PyTorch Sequential module **Topic:** Language Models & Tools **Importance:** Peripheral
265	**Keyword:** Pretrained model parameter counts **Topic:** Language Models & Tools **Importance:** Peripheral